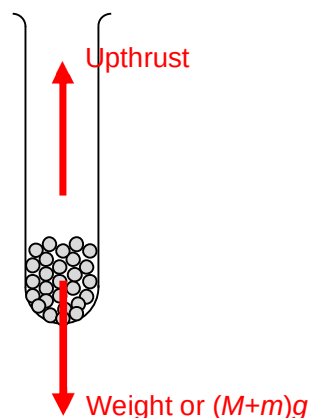


8 (a) (i)



Arrow should be equal in size (cannot be more than 1mm different in length)
Labelling should be fully spelled out for Upthrust.

- (ii) at equilibrium, Weight = $AH\rho g$
at displacement y , Upthrust = $A(H+y)\rho g$

Taking downward to be positive,

$$\text{Weight} - \text{Upthrust} = (M+m)a$$

$$AH\rho g - A(H+y)\rho g = (M+m)a$$

$$a = -\left(\frac{\rho Ag}{M+m}\right)y$$

- (iii) Using $W = AH\rho g$, $(M+m)g = AH\rho g$
 $1/H = A\rho / (M+m)$

Hence,

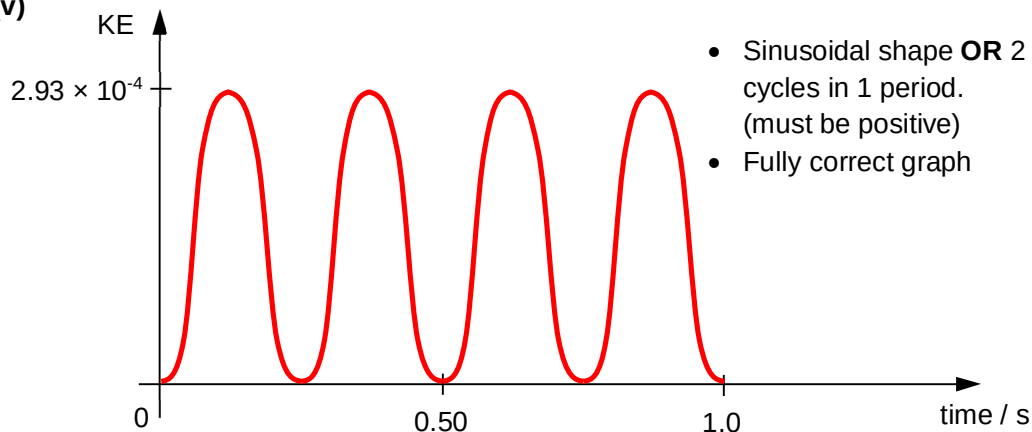
$$\omega^2 = (g/H)$$

$$(2\pi/T)^2 = (9.81/0.062)$$

$$T = 0.50$$

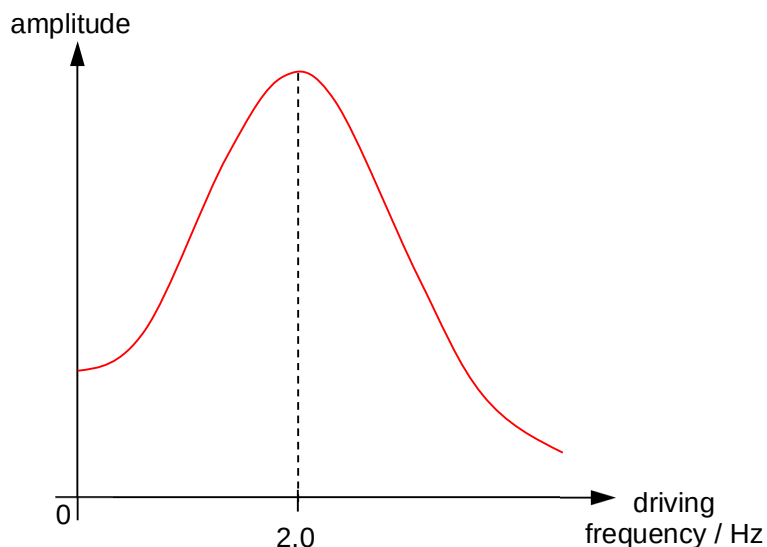
- (iv) Max KE = $\frac{1}{2}(M+m)(y^2)(\omega^2)$
= $\frac{1}{2}(0.012 + 0.025)(0.01^2)(9.81/0.062)$
= $2.93 \times 10^{-4} \text{ J}$

(v)



- (vi) As the test-tube oscillates, it experiences drag force / viscous force / dissipative forces exerted by the water. This results in light damping and energy is gradually lost as heat.

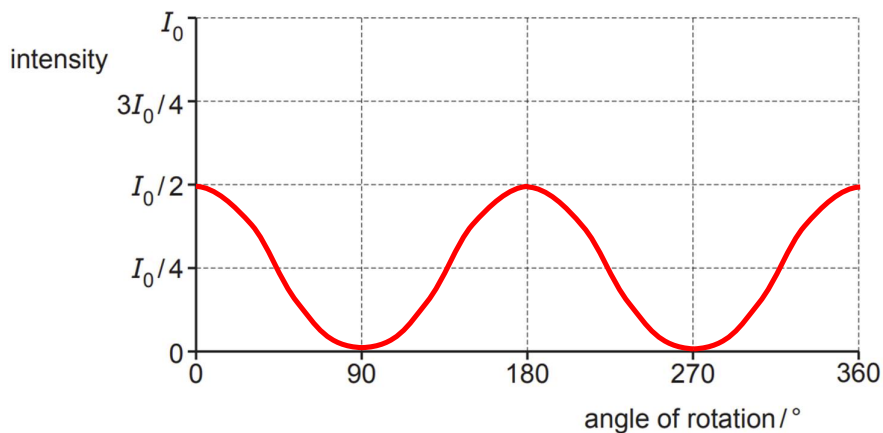
(vii) 1.



Shape + labelling of resonant frequency

2. Add more ball bearings to the test tube.
This will decrease the natural frequency of the system and hence be closer to the driving frequency.

(b) (i)



1 mark for max intensity at $0.5 I_0$

1 mark for correct shape (sine or cosine but 2 cycle)

- (ii) plane of polarisation of light from F_1 is at 45° / not perpendicular to the plane of polarisation of F_3 (and some light emerges from F_3)

plane of polarisation of lights from F_3 is at 45° / not perpendicular to the plane of polarisation of F_2

- (iii) Intensity = $(0.5 I_0)(\cos^2 45^\circ)(\cos^2 45^\circ) = 0.125 I_0$